

Numbers Are Based

Elijah White

1 Introduction

When we do math, we usually do not think about what number base we are using. Because we are taught in base-10 from a young age, we think of the decimal system as the default, even though this is not inherently true. The primary reason that we humans use base-10 is because we have 10 fingers. There are a vast number of different systems of number representation that have their own benefits and detriments. Of these, there are many types: positive and negative integer bases, non-real bases, and irrational bases are some examples.

1.1 How Number Bases Work

We use number bases to express the value of numbers. In positional number systems, we do this by writing strings of symbols called digits. Different digits have different place values. These place values depend on the base and are defined to be powers of the base.

The value of a number represented in a certain number base is found by summing the products of each digit and its respective place value. For example, 123.45 (base-10) is $1(10^2) + 2(10^1) + 3(10^0) + 4(10^{-1}) + 5(10^{-2})$.

The radix point is a dot that separates the integer place values from the fractional ones, e.g. the decimal point in base-10. For any base b , on the left of the radix point, we start with the $b^0 = 1$ place value. To the left of that is the $b^1 = b$ place value, then the b^2 place value, etc. To the right of the radix point, there is the b^{-1} place value, the b^{-2} place value, etc.

In addition to having place value, digits are restricted to certain positive integers, and for a reason. If we had a non-integer digit p in a place value b^k , we could instead represent the value pb^k as $[p]b^k + b(p - [p])b^{k-1}$, and $0 < b(p - [p])b^{k-1} < b^k$, i.e. we could have the digit in the place value b^k be $[p]$ and express the rest of the value in smaller place values. For positive bases, we do not need negative digits because, to express negative numbers, we instead use a negative sign that applies to all of the digits. For a natural number base b , digits are limited to being integers 0 to $b - 1$ (we will explain why later). For bases that exceed 10, we use other symbols, like letters, for digits. For example, base-16 uses 0 to 9 and a to f.

However, this is not the only way to represent numbers. Number systems that use bases as previously described are called positional number systems, and a digit is a type of numeral, which is any symbol used to represent a number. They are called positional because the position of a numeral determines its value. Most of the numeral systems that we use today, such as decimal, are positional number systems. Other types of number systems include additive, multiplicative, and

ciphered numeral systems. They also can have bases, but the bases have a different significance than they do in positional systems.

1.2 A Brief History of Positional Number Systems

The first civilization to develop a positional number system, and just a number system in general, that we know of were the Sumerians. They used the sexagesimal, or base-60, system. They used symbols for 1 and 10 to construct digits 1 to 59. For example, 47 would be four of the 10 symbols and seven of the 1 symbols written together. They could then express numbers in base-60 by placing these digits horizontally from right to left in ascending order of place value, like we do in decimal. So, even though they used base-60, their system had elements of base-10 in regards to how they constructed digits. Also, at first, they did not have a symbol for 0 and instead left an empty space for any place value that did not contribute any value to the target number. Circa 300 B.C.E., however, the Babylonians, who adopted the Sumerians' system, invented a symbol for 0 as a placeholder to replace this empty space (but it was only a placeholder, they did not use the symbol for the actual number 0). Base-60 is still prevalent today: there are 60 seconds in a minute, 60 minutes in an hour, and $360 = 60 \cdot 6$ degrees in a full rotation.

The Mayans had a similar system to that of the Sumerians. They used a base of 20, and they had symbols for 5 and 1 with which they constructed their digits 1 through 19. They placed these from bottom to top in ascending order of place value. After the Babylonians, around 400 C.E., they too independently invented 0 as a placeholder for place values that were empty.

The Chinese numeral system, which was invented sometime in the 2nd millennium B.C.E., is a multiplicative one. This means that each numeral consists of two symbols, when denoting the place value, and one denoting the numerical value. For example, they would use the symbols for hundreds and the symbol for 4 together to represent 400. So, a multiplicative system is similar to a positional one, but each digit consists of two numerals instead of one.

Finally, the Hindu-Arabic numeral system, which evolved from Brahmi and Gwalior numerals, was implemented and developed by Indian mathematicians by the 7th century. Its numerals replaced Roman numerals in Europe around the 12th or 13th century and later Chinese numerals in China, and they are the ones we use today.

1.3 Some Notation and Terminology

We denote what base we are using with a subscript that is the base in base-10; e.g. 57_{13} is in base-13₁₀. Assume that all numbers and arithmetic are in base-10 unless specified otherwise.

A number system (b, D) is that which uses base- b and the set D as digits. Like the base alone, the system used to express a number is denoted by a subscript: $a_{(b,D)}$ is a in base- b using digits D .

We denote the digit in the place value b^k of a number in base- b as $(d_k)_b$, or sometimes with another letter like c_k , where $k \in \mathbb{Z}$. So, we say that the value of a digit $(d_k)_b$ is $d_k b^k$ (we will always express the base, b , in base-10). A number a can be represented in base- b in base- b positional notation by $(d_s d_{s-1} \dots d_2 d_1 d_0 . d_{-1} d_{-2} \dots d_{t+1} d_t)_b$, where $a = \sum_{k=t}^s (d_k b^k)$. If a digit consists of multiple

terms or characters, it will be enclosed by parentheses and possibly separated from other digits with a space for clarity.

Note that if an equation is true in any base, it is true in all bases, as changing the base only changes how we represent values, not the values themselves.

When doing arithmetic, we “carry” numbers over to certain place values and “borrow” from certain place values. Let us just quickly make the definitions of these processes clear and consistent. Let a be the result of an arithmetic operation in base- b with the set of digits $\{d_k\}$, and let b^h be the place value that we are currently working in. When we say that we carry a number n over p places, we increase d_{h+p} by n and decrease d_h by nb^p , which preserves the value of the number because we change its value by $nb^{h+p} - nb^p(b^h) = 0$. When we borrow n from p places away, we decrease d_{h+p} by n and increase d_h by nb^p , which works because we change the value of the number by $-nb^{h+p} + nb^p(b^h) = 0$.

We say that a complex number can be expressed in a number system if that number can be represented in the number system as either, for real bases, as the sum of a real and imaginary part with no other operations, or, for non-real bases, as one number without the use of any operations.

So, we can think of a number system (b, D) as “complete” if every number can be expressed in base- b with the set of digits D in exactly one “useful” unique way. We say “useful” here because by one unique way, we mean “one unique way” like it is in the decimal system, as every number technically has two unique representations in decimal, e.g. ten is both 10 and $9.\bar{9}$.

We can think of a system as “under-complete” if it lacks the ability to express at least one number in base- b with digits D . We can think of it as “over-complete” if it can express at least one number in base- b with digits D in more than one unique way.

Let D_b denote a set of digits in base- b such that the system (b, D_b) is as complete as possible, and/or just the standard set of digits that we use for base- b .

Let $[n]$ denote the set of nonnegative integers less than n .

2 Positive Integer Bases

For a positive integer base b , $D_b = [b]$, which is why base-10 uses digits 0 to 9.

In base- b , we need the digit $b - 1$ because, for example, without it, it would not be possible to express $b - 1$ as a single number. For example, we need 9 in base-10 because without 9, we could not express 9 in base-10 without the use of operations. We would inconveniently need to write 9 as $10 - 1$ or $8 + 1$ or something similar. So, the system $(b, [b - 1])$ would be under-complete. Similarly, we need the digits $b - 2$, $b - 3$, $\dots$, and 0.

We do not need the digit b because, if we had $(d_k)_b = b$ for any k , the value of d_k would be $b(b^k) = b^{k+1}$. Therefore, $b = 10_b$, and instead of $d_k = b$, we can have $d_{k+1} = 1$. So, the system $(b, [b + 1])$ is over-complete.

Extending this, we do not any digit $(d_k)_b \geq b$, which we will put in the form $qb + r$ ($q, r \in \mathbb{N}$) where $r < b$, because then the value of d_k would be $(qb + r)b^k = qb^{k+1} + rb^k$ (we could also reason that $qb + r = (qr)_b$). So instead of $d_k = qb + r$, we could have $d_{k+1} = q$ and $d_k = r$. If $q \geq b$, then we could repeat this process with $d_{k+1} = q$.

Now, does D_b have to be $[b]$? Not necessarily. We could instead have $D_b = \{[b] - n\}$ for an integer n where $0 < n < b$, which would include negative digits. For example, we could use $[10] - 4$ or the integers -4 to 5 (inclusive) for base-10. Since the value of $(d_k)_{(b,[b])}$ is

$$d_k b^k = (b + d_k - b)b^k = 1(b^{k+1}) + (d_k - b)b^k,$$

to express the value of $(d_k)_{(b,[b])}$ in $(b, [b] - n)$ where $d_k \geq b - n \notin [b] - n$, we can have $(d_{k+1})_{(b,[b]-n)} = 1$ and $(d_k)_{(b,[b]-n)} = (d_k)_{(b,[b])} - b < 0$. More simply put, $(d_k)_{(b,[b])} = (1(d_k - b))_{(b,[b]-n)}$. For example, if we wanted to write 76 in the system $(10, [10] - 4)$, i.e. using the digits -4 to 6 :

$$76_{(10,[10])} = 1(10^2) - 2(10^1) - 4(10^0), \text{ which is } (1(-2)(-4))_{(10,[10]-4)}.$$

However, the standard is $D_b = [b]$ because nonnegative digits are simpler to work with.

To count in positive bases other than 10, instead of converting every number from base-10 to base- b , you can count through base-10 and only record numbers that only use the digits $0, 1, \dots, b-1$, or, better yet, think of the numbers that only use these digits in increasing order from 1. For example, to count from 1 in base-3:

$$1, 2, 10, 11, 12, 20, 21, 22, 100, 101, 102, 110, 111, 112, 120, 121 \dots$$

Also, unary, or base-1, only uses one digit, 0. To make unary at least somewhat usable, we can think of each “0” representing a 1 because 1 to any integer power is 1, and so unary is essentially a system of tally marks.

2.1 Conversion

The easiest way to convert a number a from base- b to base-10 is to sum the values of the digits of a using base-10 arithmetic, which is what we are used to:

$$a_b = (d_s \dots d_2 d_1 d_0 . d_{-1} d_{-2} \dots d_t)_b \Rightarrow a = \sum_{k=t}^s (d_k b^k)$$

We will discuss ways of converting from base-10 to other bases and between two non-decimal bases. There are several methods to convert a from base-10 into base- b , where $b \in \mathbb{N}$. One such method is the following process:

1. Find the largest power of b , b^N , such that $b^N \leq a$ (use base-10).
2. Let d_N (the digit in the place value b^N) be $\lfloor \frac{a}{b^N} \rfloor$, and let R_N be the remainder of $\frac{a}{b^N}$ (again, use base-10 for this arithmetic).
3. Let $d_{N-1} = \lfloor \frac{R_N}{b^{N-1}} \rfloor$, and let R_{N-1} be the remainder of $\frac{R_N}{b^{N-1}}$ (use base-10).
4. Repeat step 3 with d_{N-2} and R_{N-2} , d_{N-3} and R_{N-3} , etc. n times until

$$a - \sum_{k=N-n+1}^N (d_k b^k) = 0, \text{ i.e. } R_{N-n+1} = 0.$$

5. If $N - n + 1 > 0$, which means that we have finished expressing all of a 's value before reaching all the way to the place value b^0 , concatenate $N - n$ zeros at the end of the result from the previous steps.

In other words, starting at the largest power of b less than or equal to a , we repeatedly find the largest multiple of a power of b , $d_k b^k$, less than or equal to the remaining amount of a , make d_k the digit for the b^k place value, and subtract $d_k b^k$ from a .

This power method starts by finding the digit in the largest place value. There is another method, though, that starts at the smallest place value. Let a be the number we are converting and we will start at the place value b^s (use base-10 for the arithmetic of every step):

1. Let d_s be the remainder of $\frac{a}{b^{s+1}}$, divided by b^s , and let $R_0 = a - d_s b^s$.
2. Let d_{s+1} be the remainder of $\frac{R_0}{b^{s+2}}$, divided by b^{s+1} , and let $R_1 = R_0 - d_{s+1} b^{s+1}$.
3. Let d_{s+2} be the remainder of $\frac{R_1}{b^{s+3}}$, divided by b^{s+2} , and let $R_2 = R_1 - d_{s+2} b^{s+2}$.
4. Continue this process with d_3 and R_3 , d_4 and R_4 , etc. n times until

$$a - \sum_{k=s}^{n-1} (d_k b^k) = 0, \text{ i.e. } R_{n-1} = 0$$

Said in English, we are finding the remainder of the number divided by the next place value, which is the value that we need to express in the current place value (since it is smaller than the next place value and cannot be expressed in any larger place values), and then setting the digit equal to that divided by the current place value. Then, we subtract that digit's value from the number and do the same with the next place value. For example, if we wanted to convert 14 into base-3, we would first find the remainder of 14 divided by 3, which is 2, to find the value that we need to express in the ones digit because everything else can be expressed in larger place values, like those of 3 and 9.

This method only works well for real bases, though, as calculating remainders is different when dealing with imaginary numbers.

If a is an integer, and we do not already know the best place value to start at, it is usually the easiest to start with $s = 0$. However, if the number we are converting is not an integer, it might be difficult to find the place value to start at. If we start at the wrong place value, this method will not work because we are assuming that the values of all d_k for $k < s$ are 0.

We can use this to convert non-integer values by guessing and checking with different starting bases, but this might not work well if the radix representation of a number in the other base does not terminate. If this is the case, we can use division to convert it by writing it as a fraction, converting the numerator and denominator, and then performing division. To get an approximation in another base when the number is irrational, we can use a base-10 approximation for it and then convert that. So, if we wanted an approximation of π in base-7, we could convert 3.14. Note that since the representation of the number π does not use a positional number system (we just use the symbol π), π , and other numbers that we use special characters for, mean the same thing in all bases, kind of like 0.

As an example, say we want to convert 197 into base-12. Using the power method:

1. The greatest power of 12 that does not exceed 197 is $12^2 = 144$.
2. d_2 is $\lfloor \frac{197}{144} \rfloor$, which is 1.
3. d_1 is $\lfloor \frac{197-1(144)}{12} \rfloor = \lfloor \frac{53}{12} \rfloor$, which is 4.
4. d_0 is $\lfloor \frac{53-4(12)}{1} \rfloor$, which is 5.
5. So, $197 = 145_{12}$.

Using the remainder method:

1. d_0 is the remainder of $\frac{197}{12}$, which is 5.
2. d_1 is the remainder of $\frac{197-5}{144} = \frac{192}{144}$, which is 48, divided by 12, which is 4.
3. d_2 is the remainder of $\frac{192-4(12)}{12^3}$, which is 144, divided by 144, which is 1.
4. So, like before, $197 = 145_{12}$.

We can use a very similar method to convert directly from any base- c to any other base b . The only difference when converting from a non-decimal base c is that we have to do arithmetic in base- c , like how we did all arithmetic in base-10 when converting from base-10. We would also have to first convert the base b itself, which is given in base-10, into base- c . However, it is most likely faster to convert from base- c to base-10 and then from base-10 to base- b , as arithmetic in other bases takes much longer and is much more difficult for us.

Even so, there is a method of grouping for converting between bases that are powers of each other. To convert a number a from base- b to base- b^h where h is a positive integer, we can take the digits of a_b h at a time starting at the radix point to the left and right. We can then convert each of those groups of h digits into single digits in base- b^h and concatenate all the results in order of how the groups of digits were formed to get a_{b^h} , including the radix point. The reason why this works is a consequence of the fact that $(n-1)(n^0 + n^1 + n^2 + \dots + n^{k-1} + n^k) = n^{k+1} - 1$. Rewriting that for this context:

$$(b-1)(b^0 + b^1 + b^2 + \dots + b^{h-2} + b^{h-1}) = b^h - 1.$$

Since $(b-1)(b^0 + b^1 + b^2 + \dots + b^{h-2} + b^{h-1})$ is the maximum value that the groups of digits can obtain (since $b-1$ is the largest digit possible in base- b), the range of possible values that the groups of digits produce is the same as the range of digits of b^h . For example, the maximum possible value of such a group of 3 digits in base-4 is $3(4^0 + 4^1 + 4^2) = 63 = 4^3 - 1$, which is equal to the maximum possible value of a digit in base-64.

So, as an example of this method of conversion, 12032.2_4 is equal to $6e.2_{64}$ because $12_4 = 6_{64}$, $032_4 = e_{64}$, and $2_4 = 2_{64}$.

To convert from base- b^h to base- b , reverse this process by converting each digit in base- b^h into a sequence of h digits in base- b .

2.2 Arithmetic

To do arithmetic in different bases, we can just convert the operands into base-10, do the arithmetic in base-10, and convert the result back into the original base. However, this is not particularly interesting, and we will discuss how to do arithmetic in different bases without converting to base-10.

Let the k th place value be the one that we are working in, and let m_k and n_k be the digits of the operands in the k th place value. Assume that m_k is a digit of the minuend and n_k is a digit of the subtrahend for subtraction.

When doing arithmetic in different bases, there are two main differences: when to and what to carry and borrow.

Addition is very similar for all positive integer bases. For addition in a positive integer base b , we carry a 1 over one place when $m_k + n_k \geq b$, since $b(b^k) = b^{k+1}$.

Subtraction is also very similar for all positive integer bases. We borrow b from one place away when $m_k - n_k \leq 0$. For example, $21_6 - 5_6 = 12_6$.

Multiplication is also consistent. When $m_k n_k \geq b$, we carry the b s digit of $m_k n_k$ over 1 place (like how we would carry the 10s digit of the product over 1 place). So, one-digit times one-digit multiplication works differently in every base. For example, $3_6 \times 4_6 = 20_6$.

The only difference in division is the process of determining how many of the divisor “go in to” the specified digits in the dividend.

Doing operations in non-decimal bases (especially multiplication and division) can be difficult since we do not have non-decimal multiplication tables memorized, so having some sort of multiplication table can help with these operations, like this one for base-12:

x	1	2	3	4	5	6	7	8	9	χ	ϵ	10
1	1	2	3	4	5	6	7	8	9	χ	ϵ	10
2	2	4	6	8	χ	10	12	14	16	18	1χ	20
3	3	6	9	10	13	16	19	20	23	26	29	30
4	4	8	10	14	18	20	24	28	30	34	38	40
5	5	χ	13	18	21	26	2ϵ	34	39	42	47	50
6	6	10	16	20	26	30	36	40	46	50	56	60
7	7	12	19	24	2ϵ	36	41	48	53	5χ	65	70
8	8	14	20	28	34	40	48	54	60	68	74	80
9	9	16	23	30	39	46	53	60	69	76	83	90
χ	χ	18	26	34	42	50	5χ	68	76	84	92	$\chi 0$
ϵ	ϵ	1χ	29	38	47	56	65	74	83	92	$\chi 1$	$\epsilon 0$
10	10	20	30	40	50	60	70	80	90	$\chi 0$	$\epsilon 0$	100

Figure 1: Duodecimal times table

where χ is the digit for 10 and ϵ is the digit for 11.

As mentioned previously, we can use division to convert non-integer values. For example, if we wanted to convert $\frac{1}{7}$ into base-12, we would convert the numerator and denominator into base-12, which gives us the same $\frac{1}{7}$, and then we would perform long division (assume all arithmetic and numbers are in base-12):

$$\begin{array}{r}
0.\overline{186\chi} \\
7 \overline{) 1.0000} \\
\underline{-7} \quad 7 \text{ goes into } 10: 1 \text{ time} \\
50 \quad 10 - 7 = 5 \\
\underline{-48} \quad 7 \text{ goes into } 50: 8 \text{ times, and } 7 \times 8 = 48 \\
40 \quad 50 - 48 = 4 \\
\underline{-36} \quad 7 \text{ goes into } 40: 6 \text{ times, and } 7 \times 6 = 36 \\
60 \quad 40 - 36 = 6 \\
\underline{-5\chi} \quad 7 \text{ goes into } 60: \chi \text{ times, and } 7 \times \chi = 5\chi \\
1 \quad 60 - 5\chi = 1, \text{ and we are now back to where we started}
\end{array}$$

So, $\frac{1}{7} = 0.\overline{186\chi}_{12}$.

3 Negative Integer Bases

Similarly to positive bases, for a negative integer base $-b$ ($b \in \mathbb{Z}$ and $b > 0$), $D_b = [b]$.

We once again do not need a digit of b because if we had $(d_k)_b = b$, d_k has value

$$b(-b)^k = -(-b)(-b)^k = -(-b)^{k+1}.$$

This can be rewritten as

$$\begin{aligned}
(-b)^{k+2} - (-b)^{k+2} - (-b)^{k+1} &= (-b)^{k+2} - (-b)(-b)^{k+1} - (-b)^{k+1} \\
&= (-b)^{k+2} + b(-b)^{k+1} - (-b)^{k+1} \\
&= (-b)^{k+2} + (b-1)(-b)^{k+1} \\
&= 1(-b)^{k+2} + (b-1)(-b)^{k+1} + 0b^k
\end{aligned}$$

This means that $b = (1 \ (b-1) \ 0)_{-b}$, and instead of expressing a number using b as d_k , we can have $d_{k+2} = 1$ and $d_{k+1} = b-1$. For example, $10 = 190_{-10}$.

Extending this, the value of a digit $(d_k)_b = qb + r \geq b$ ($q, r \in \mathbb{N}, r < b$) would be

$$\begin{aligned}
(qb + r)(-b)^k &= qb(-b)^k + r(-b)^k \\
&= -q(-b)^{k+1} + r(-b)^k \\
&= (-b)^{k+2} - (-b)^{k+2} - q(-b)^{k+1} + r(-b)^k \\
&= (-b)^{k+2} + b(-b)^{k+1} - q(-b)^{k+1} + r(-b)^k \\
&= 1(-b)^{k+2} + (b-q)(-b)^{k+1} + r(-b)^k
\end{aligned}$$

So, $qb + r = (1 \ (b-q) \ r)_b$, and we could have $d_{k+2} = 1$, $d_{k+1} = b-q$, and $d_k = r$, e.g. $34 = 174_{-10}$. But what if $b-q < 0$?

We do not need negative digits in negative bases because if $(d_k)_b = -n$ where $n \in \mathbb{N}$ and $n < b$, d_k has value

$$\begin{aligned} -n(-b)^k &= (-b + b - n)(-b)^k \\ &= (-b)(-b)^k + (b - n)(-b)^k \\ &= 1(-b)^{k+1} + (b - n)(-b)^k \end{aligned}$$

So, $-n = (1(b - n))_{-b}$, and instead of $d_k = -n$, we can have $d_{k+1} = 1$ and $d_k = b - n$ to express the same value, e.g. $-7 = 13_{-10}$. More generally, if $d_k = -(qb + r) \leq -b$, where $q, r \in \mathbb{Z}$, $r < b$:

$$\begin{aligned} -(qb + r)(-b)^k &= q(-b)^{k+1} - r(-b)^k \\ &= q(-b)^{k+1} + 1(-b)^{k+1} + (b - r)(-b)^k \\ &= (q + 1)(-b)^{k+1} + (b - r)(-b)^k \end{aligned}$$

So, $d_k = -(qb + r)$ can be replaced with $((q + 1)(b - r))_{-b}$.

An interesting feature of negative bases is that they can represent negative numbers without using a negative sign, since they have place values that are negative. For example, $-13 = 27_{-10}$.

The counting process in negative bases is much different than that of positive bases. It is less straightforward. For example, in negadecimal, or base- (-10) , counting from 1 is as follows:

$$1, 2, \dots, 9, 190, 191, 192, \dots, 199, 180, 181, \dots$$

Negative bases also have what is called a *clearing algorithm*, which is something that helps us make arithmetic and conversion in negative bases easier. It allows us to temporarily ignore the restrictions on the digits that we can use. For a base $-b$, this is accomplished by adding b to any digit d_k and then adding 1 to d_{k+1} . This works because $1(-b)^{k+1} + b(-b)^k$ is always 0. So, we can also add and subtract multiples of 1 and b .

This means that we can temporarily have digits that are not in $D_{-b} = [b]$ and then “clear” them by cleverly adding and subtracting the polynomial $1(-b)^{k+1} + b(-b)^k$. We will discuss the clearing algorithm’s application more in the next section.

We can also use this idea for positive integer bases, although it is not as useful because, per what we will later see, it would not make anything any easier: it would just change the order of steps for something like addition or multiplication. The clearing algorithm for a positive base b would be adding/subtracting $-b$ to d_k and then 1 to d_{k+1} .

3.1 Conversion

The power method that we discussed for positive bases does not work for negative bases because we cannot really determine the largest multiple of a power of the base that goes into a number when some of the place values are negative.

The remainder method, however, does not rely on the second base being positive since it uses remainders to find the digits. So it also works for negative bases, with one extra step:

5. If the base is negative, we may have gotten negative digits in the previous steps. If so, make them positive by borrowing a -1 from one place value away, that is increase the next digit by 1 and increase the current digit by b (base is $-b$).

This step is one of the uses of the clearing algorithm.

As an example that the remainder method works for negative bases too, let us convert 16 into base- (-3) :

1. d_0 is the remainder of $\frac{16}{-3}$, which is 1.
2. d_1 is the remainder of $\frac{16-1}{(-3)^2} = \frac{15}{9}$, which is 6, divided by -3 , which is -2 .
3. d_2 is the remainder of $\frac{15-(-2)(-3)}{(-3)^3} = \frac{9}{-27}$, which is 9, divided by $(-3)^2 = 9$, which is 1.
4. So, we have $16 = 1(-2)1_{-3}$. But we do not want negative digits, so we can increase the 9s place value by 1 and increase the -3 s place value by 3.
5. So, $16 = 211_{-3}$.

We can check this: $2(-3)^2 + 1(-3) + 1 = 18 - 3 + 1 = 16$.

However, instead of using remainders, we could also use only the clearing algorithm, which, for base- (-3) , is adding multiples of 13_{-3} from a place value:

$$\begin{array}{r}
 16 \text{ is the number we are converting} \\
 -5 - 15 = -5(1(-3)^1 + 3(-3)^0) = 0 \\
 \hline
 \begin{array}{r}
 +2 \quad +6 \\
 2 \quad 1 \quad 1
 \end{array} = 2(1(-3)^2 + 3(-3)^1) = 0
 \end{array}$$

Using only the clearing algorithm would take a long time for larger numbers, though.

To convert non-integer values, we would use division, as with positive bases.

3.2 Arithmetic

In addition for a negative integer base $-b$, when $m_k + n_k \geq b$, we carry a -1 over one place. This works because, as previously discussed, $b(-b)^k = -1(-b)^{k+1}$. For example, $(57 + 284)_{-10}$:

$$\begin{array}{r}
 -1 \\
 284 \\
 + 57 \\
 \hline
 121
 \end{array}$$

In subtraction, when $m_k - n_k = 0$, we borrow a -1 from 1 place away (we increase the digit in the next largest place by 1 and increase the current digit by b), because $1(-b)^{k+1} + b(-b)^k = 0$. However, subtraction is equivalent to adding the negative of a number. So, if we wanted to, instead of subtracting a number n_{-b} from m_{-b} , we could find the negative of n in base- $(-b)$ and then add m_{-b} and $-n_{-b}$.

In multiplication, when $m_k n_k = qb + r \geq b$ ($q, r \in \mathbb{N}$ and $q, r < b$), we carry a 1 over two places and a $b - q$ over one place. This is because, as seen previously, $qb + r = (1(b - q)r)_b$. This gives us a way to find the negative of a number in base- $(-b)$ without converting to base- b and then negating it and converting back. Instead, we can multiply by $(1(b - 1))_{-b} = 1(-b) + (b - 1) = -1$. For example if we wanted to find the negative of 152_{-10} , we could multiply by $19_{-10} = -1$:

$$\begin{array}{r}
-4-1 \\
152 \\
\times \quad 19 \\
\hline
548 \quad = (152 \times 9)_{-10} \\
+ \quad 1520 \quad = (152 \times 10)_{-10} \\
\hline
0068 \\
-1
\end{array}$$

So, $(-152 = 68)_{-10}$. Notice that when we move on to multiplying by the digit in the next biggest place value, we place a zero and then multiply by that digit, just like we do in positive number bases. This is because, no matter the base, multiplying by something that is the base times larger shifts the result to the left by one place.

Division is similar to that for non-decimal positive integers in that we go one digit at a time from the greatest place value finding how many of the divisor go into that part of the dividend and then subtracting to find the remainder. But it is different in that sometimes we will get negative values for a digit, which leads to “digits” that are more than one digit long that we have add together at the end, like the 18_{-10} in this example of $(427 \div 3)_{-10}$:

$$\begin{array}{r} 1 \\ 189 \\ 3 \overline{)427} \\ \underline{-3} \quad 3 \text{ goes into 4: 1 time} \\ 12 \\ \underline{-14} \quad 3 \text{ goes into } 12_{-10} = -8: -2 = 18_{-10} \text{ times} \\ 187 \\ \underline{-187} \quad 3 \text{ goes into } 187_{-10} = 27: 9 \text{ times} \\ 0 \end{array}$$

So, $(427 \div 3)_{-10} = 289_{-10}$. We can check this by doing $(3 \times 289)_{-10}$:

$$\begin{array}{r} -2-2 \\ 289 \\ \times \quad 3 \\ \hline 427 \end{array}$$

There is one more thing. Instead of carrying a -1 over 1 place for addition like we have done, there is an alternative of carrying a 1 over 2 places and a $b - q$ over 1 place, where q is the b s digit of the sum of digits in a column. But, this is really only useful when the sum in the next highest place is 0, since we do not want to add a -1 to it. Even in this case, though, we can use the borrowing that we did for subtraction of borrowing a 1 from 1 place away from the -1 to increase the -1 by

b , which ultimately leads to the same thing as what we would get by just carrying -1 .

Also, if we do always use this method of carrying, it creates many roundabout scenarios. For example, the $(57 + 284)_{-10}$ that we did earlier:

$$\begin{array}{r}
 \dots \\
 19 \\
 19 \\
 18 \\
 19 \\
 284 \\
 + \underline{57} \\
 \dots 00121
 \end{array}$$

This carrying method can also create infinite loops of carrying, like with the example above and with $(1 + 19)_{-10}$:

$$\begin{array}{r}
 \dots \\
 19 \\
 19 \\
 19 \\
 + \underline{1} \\
 \dots 000
 \end{array}$$

4 Pure Imaginary Bases

Non-real bases have many interesting properties, such as their ability to represent all complex numbers in one part and with no sign.

For a pure imaginary base bi , instead of $D_{bi} = [b]$, $D_{bi} = [b^2]$. This is because every pure imaginary base bi has an effective negative base $(bi)^2 = -b^2$. The effective negative base determines the allowable digits: base- bi uses the same digits as base- $(-b^2)$.

All place values $(bi)^k$ where k is even are real. The effective negative base is the square of bi because all of the real place values will be powers of $(bi)^2 = -b^2$. This means that real part of any complex number is essentially expressed in base- $(-b^2)$. The only difference is that the real place values are dispersed among imaginary place values. So we need the same digits in base- bi as in base- $(-b^2)$ to be able to express all real numbers.

All place values $(bi)^k$ where k is odd are pure imaginary numbers. All such place values can be rewritten as $bi(bi)^{k-1} = bi(-b^2)^{\frac{k-1}{2}}$. So, all of the imaginary place values are the products of bi and an integer power of the effective negative base. So, to express any pure imaginary number yi in base- bi , we can first find what $\frac{yi}{bi} = \frac{y}{b}$, a real number, is in base- (bi) and then shift that to the left by one place value (essentially multiplying it by bi) to undo the division by bi . Therefore, we need

the same digits as the effective negative base to be able to express all pure imaginary numbers. So,

$$\begin{aligned} a_{bi} &= (d_{2s} \dots d_3 d_2 d_1 d_0 . d_{-1} d_{-2} d_{-3} \dots d_{2t})_{bi} \\ &= (d_{2s} \dots d_2 d_0 . d_{-2} \dots d_{2t})_{-b^2} + bi \cdot (d_{2s-1} \dots d_3 d_1 . d_{-1} d_{-3} \dots d_{2t+1})_{-b^2}. \end{aligned}$$

Together we need only the digits of the effective negative base to have a complete system.

One pure imaginary base is base- $2i$, or quater-imaginary. Its effective negative base is $-2^2 = -4$, and $D_{2i} = 0, 1, 2, 3$. Some of its place values are

$$\dots (2i)^5 = 32i, (2i)^4 = 16, (2i)^3 = -8i, (2i)^2 = -4, (2i)^1 = 2i, (2i)^0 = 1 \dots$$

$1, -4, 16, \dots$ are powers of -4 and therefore place values of base- (-4) .

$2i, -8i, 32i, \dots$ are products of $2i$ and powers of -4 .

But, how would we represent any pure imaginary number hi with an odd coefficient h in base- $2i$ if all imaginary place values have even coefficients? If we could find a way to express i or $-i$ in base- $2i$, then we could express any such hi . Luckily, there is one: $0.2_{2i} = -i$, since

$$0.2_{2i} = 2(2i)^{-1} = \frac{2}{2i} = \frac{1}{i} = \frac{1 \cdot i}{i \cdot i} = \frac{i}{-1} = -i.$$

So, for example, $i = 10.2_{2i}$ and $3i = 20.2_{2i}$.

Similarly, with any other pure imaginary base bi , d_{-1} can be used to express pure imaginary numbers that are not multiples of b .

As with any other bases, we can express fractional values. For example, $0.5 = \frac{i}{2i} = \frac{10.2_{2i}}{2i} = 1.02_{2i}$ (dividing by $2i$ shifts the numerator 10.2_{2i} one place value to the right).

Like negative bases, pure imaginary bases have a clearing algorithm. We can add/subtract multiples of $(1 \ 0 \ b^2)_{bi}$ since $1(bi)^2 + 0(bi) + b^2 = -b^2 + b^2 = 0$.

4.1 Conversion

To convert a number $x + yi$ from base-10 into an imaginary base bi :

1. Take the real part x and convert it into base- $(-b^2)$ using the remainder method or a clearing algorithm (or, if the numbers are small, we can use intuition on some level) to get $(\dots d_2 d_1 d_0 . d_{-1} d_{-2} \dots)_{-b^2}$, which are the digits for the real place values.
2. Take the imaginary part y , divide it by 2 and convert that into base- $(-b^2)$ using the remainder method to get $(\dots c_2 c_1 c_0 . c_{-1} c_{-2} \dots)_{-b^2}$, which are the digits for the imaginary place values.
3. We combine these to get $\dots c_2 d_2 c_1 d_1 c_0 d_0 . c_{-1} d_{-1} c_{-2} d_{-2} \dots$, which is $(x + yi)_{bi}$.

For example, let us convert $19 - 7i$ into base- $2i$. The effective negative base is -4 :

1. 19 in base- (-4) is $103_{-4} = 1(-4)^2 + 3(-4)^0 = 16 + 3$.
2. $\frac{-7}{2} = -3.5$ in base- (-4) is $11.2_{-4} = 1(-4)^1 + 1(-4)^0 + 2(-4)^{-1} = -4 + 1 + 2(-\frac{1}{4})$.
3. Put these together to get 11013.2_{2i} .

Just to check, this is $16 - 8i + 2i + 3 - i = 19 - 7i$.

4.2 Arithmetic

Arithmetic is very similar in pure imaginary bases as it is in negative bases because of its possession of an effective negative base. So, we can treat arithmetic in a pure imaginary base bi as two separate arithmetic processes in base- $(-b^2)$, one in real place values and one in imaginary place values.

So, for addition, we can carry a -1 over two places, doubling the number of places we carry over for negative bases so that we “skip” over the place values not similar to the current one in terms of being real or imaginary.

In subtraction, we borrow a -1 from 2 places away, or, as with negative bases, we can find the negative of the subtrahend in base- bi and then add that to the minuend.

In multiplication, when $m_k n_k = qb + r \geq b$, we carry a 1 over four places and a $b - q$ over two places.

Division by a real number in a pure imaginary base is similar to that in a negative base, but, with non-real bases, we must determine both how many of the divisor go into the real part of some part of a number and how many go into the imaginary part. For example, $(4 + 3i) \div 2 = 2 + i \text{ R } 1$. Here is an example of $(33 \div 2)_{2i}$:

$$\begin{array}{r}
 10.2 \\
 11.2 \\
 1 \\
 2 \overline{) 33.0} \\
 \underline{-2} \quad 2 \text{ goes into } 3: 1 \text{ time, and } 2 \times 1 = 2 \\
 13 \\
 \underline{-12} \quad 2 \text{ goes into } 13_{2i} = 3 + 2i: 11.2_{2i} = 1 + i \text{ times, and } (2 \times 11.2 = 12)_{2i} \\
 10 \\
 \underline{-10} \quad 2 \text{ goes into } 10_{2i} = 2i: 10.2_{2i} = i \text{ times, and } (2 \times 10.2 = 10)_{2i} \\
 0
 \end{array}$$

So, $(33 \div 2 = 22.22)_{2i}$. We can check this in base-10:

$$\begin{aligned}
 (2 \times 22.22)_{2i} &= 2 \times (2(2i) + 2(1) + 2(-i) + 2(-\frac{1}{2})) = 2 \times (3i + 1.5) = 6i + 3, \text{ and} \\
 33_{2i} &= 3(2i) + 3 = 6i + 3
 \end{aligned}$$

If we want to divide by a pure imaginary number, we can divide after multiplying both the divisor and dividend by bi to make the divisor a real number.

To divide by a complex number, we can use a strategy that we use in base-10 to divide by complex numbers. We can multiply both the divisor and dividend by the complex conjugate of the divisor so that the divisor becomes a real number. For example, to do

$$(110023.2 \div 13)_{2i} = (19 - 9i) \div (3 + 2i),$$

we can first find the complex conjugate of 13_{2i} :

$$(\overline{13} = (-1) 3 = 1033)_{2i}$$

Next, we multiply the divisor by its conjugate:

We then multiply the dividend by the divisor's conjugate, which yields

We can check if 1023.2_{2i} is the correct answer to $(11003.2 \div 13)_{2i}$ in base-10:

$$\begin{aligned}
1103.2_{2i} &= 19 - 9i \\
13_{2i} &= 3 + 2i \\
1023.2_{2i} &= 3 - 5i \\
\frac{19 - 9i}{3 + 2i} &= \frac{(19 - 9i)(3 - 2i)}{(3 + 2i)(3 - 2i)} = \frac{39 - 65i}{13} = 3 - 5i
\end{aligned}$$

One last thing: since we use a very similar method of carrying in pure imaginary bases as in negative bases, the same points discussed about the alternative method of carrying applies to pure imaginary bases: it is helpful in some situations but can lead to infinite loops of carrying.

5 Complex Bases

Complex bases can be extremely confusing to our decimal-oriented minds, even more so than negative and pure imaginary bases.

It has been proven that any complex base system with a base of the form $-n \pm i$ where $n \in \mathbb{N}$, and only complex base systems with bases of this form, can uniquely express every Gaussian integer using $D = [n^2 + 1]$ (Káta and Szabó, 1979).

One such base is base- $(-1 + i)$, one of the simplest complex bases, which uses $D = [1^2 + 1] = [2] = \{0, 1\}$.

Like negative and pure imaginary bases, complex bases have clearing algorithms. The clearing algorithm of a base $-n \pm i$ is adding or subtracting multiples of $n^2 + 1$ from any digit d_k , then $2n$ from d_{k+1} , and 1 from d_{k+2} . This works because, after expanding and combining like terms, we can see that the polynomial

$$1(-n \pm i)^{k+2} + 2n(-n \pm i)^{k+1} + (n^2 + 1)(-n \pm i)^k = (-n \pm i)^k (1(-n \pm i)^2 + 2n(-n \pm i) + (n^2 + 1))$$

is always equal to 0, i.e.

$$(1 \ (2n) \ (n^2 + 1))_{-n \pm i} = 0.$$

5.1 Conversion

Any Gaussian integer $s + ti$ can be cleverly rewritten as

$$t(-n + i) + (s + nt), \text{ or } (t \ (s + nt))_{-n+i}$$

for a base $-n + i$, or as

$$-t(-n - i) + (s - nt) = (-t \ (s - nt))_{-n-i}$$

for a base $-n - i$.

We can then convert this form into base- $(-n \pm i)$ by using its clearing algorithm to obtain only digits that are in $D_{-n \pm i}$. Here is an example of converting $5 + 6i$ into base- $(-3 + i)$, in which we use the fact that $5 + 6i = 6(-3 + i) + (5 + 3 \times 6) = (6 \ (23))_{-3+i}$ and that the clearing algorithm for base- $(-3 + i)$ is adding multiples of $(1 \ 6 \ 10)_{-3+i}$:

So, for addition, when $m_k + n_k \geq n^2 + 1$, we carry a 1 over 3 places, a $2n - 1$ over 2 places, and a $(n - 1)^2$ over 1 place. However, if the digits to the left are large enough, we can instead carry a -1 over 2 places and a $-2n$ over 1 place, leaving the current digit at 0. This is because $((-1) (-2n) 0)_{-n \pm i} = 0$ (which can be seen if we stopped one line earlier in our use of the clearing algorithm in the previous paragraph).

Similarly to pure imaginary and negative bases, if we only ever carry using $(1 (2n - 1) (n - 1)^2 0)_{-n \pm i}$, and we never use $((-1) (-2n) 0)_{-n \pm i}$, there are situations in which we might encounter an infinite loop of carrying and/or borrowing values when doing arithmetic. For example, shown here is $(169 + 1)_{-3+i} = -1 + 1$ (Gilbert, 1984).

$$\begin{array}{r}
 169 \\
 + \quad 1 \\
 \hline
 \dots 0000000 \\
 \hline
 154 \\
 \swarrow 154 \\
 154 \\
 \swarrow 154 \\
 154 \\
 \swarrow 154 \\
 154 \\
 \swarrow 154 \\
 \dots
 \end{array}$$

Figure 2: Infinite addition carrying loop

Here, since the first three columns sum to $10 = 3^2 + 1$, we repeatedly carry a 1 over 3 places, a $2(3) - 1 = 5$ over 2 places, and a $(3 - 1)^2 = 4$ over 1 place, which leads to an infinite string of zeros. Instead, we could carry a -1 over 2 places and a $-2(3) = -6$ over 1 place, which would immediately give us 0 as the answer.

For subtraction, when $m_k - n_k = -a < 0$ ($a \in \mathbb{N}$), we increase the digit two places higher by 1, increase the digit one place higher by $2n$, and the current digit by $n^2 + 1$, making it $n^2 + 1 - a$. This is just using the clearing algorithm. If $2n > n^2$ (which is only for $n = 1$), we can do what we did for when a digit is $n^2 + 1$. If $n^2 + 1 - a < 0$, then we can repeat the process. Like with negative and pure imaginary bases, we can also avoid subtracting by multiplying the subtrahend by -1 and then adding the result to the minuend.

For multiplication, we can use a combination of polynomial multiplication and the clearing algorithm. The polynomial multiplication step emphasizes that the factors are polynomials in terms of the base $-n \pm i$, and so we multiply them how we would any other polynomials in x . Figure 3 shows the multiplication of 182_{-3+i} and 38_{-3+i} (Gilbert, 1984). The digits in this result are what we want because $D_{-3+i} = [9]$. So, $(182 \times 38 = 13546)_{-3+i}$.

We can approach division for complex bases very similarly to how we approached division for pure imaginary bases. We multiply the dividend and divisor to make sure that the divisor is real. Then, we divide by determining how many of the divisor go into both the real and imaginary parts of the digits as we work our way from the left to the right.

		1	8	2	
×			3	8	
		8	64	16	
	3	24	6		
	3	32	70	16	
		-1	-6	-10	
	-6	-36	-60		
	1	6	10		
	1	3	5	4	6

}

polynomial
multiplication

}

clearing
algorithm

Figure 3: Complex base multiplication example

Here is the short example of $(133040 \div 3)_{-2+i}$ (assume all numbers without a subscript are in base- $(-2 + i)$):

$$\begin{array}{rcl}
 1330 & = & (1 + 2i)_{10} \\
 3 \overline{) 133040} & = & (3 + 6i)_{10} \\
 \underline{-13304} & & 3 \text{ goes into } 133040 = -3i_{10}: 133 = (-i)_{10} \text{ times, and } 3 \times 133 = 13304 \\
 0 & &
 \end{array}$$

There are other ways of approaching division in complex bases. In one such approach, each complex base has a special set of remainders that it can have during division. This set of complex remainders also changes depending on the divisor and is difficult to find. If you wish to read more about this approach, I recommend reading William Gilbert's article (Gilbert, 1984, pp. 80-81).

6 Fractional and Irrational Bases

There are also real bases that are not integers.

6.1 Less Than One

The expression of any number in a base $\frac{1}{b}$ where $\frac{1}{b} < 1$ can be considered the reflection of that number's representation in base- b over the ones place value because $(\frac{1}{b})^n = b^{-n}$. So, any such base $\frac{1}{b}$ uses the same digits as base- b : $D_{\frac{1}{b}} = [b]$. For example, $18054.39_{10} = 934.5081_{\frac{1}{10}}$.

6.2 More Than One

For a base b where $b \notin \mathbb{Z}$ and $b > 1$, $D_b = [b]$. This is for very similar reasons for the limits on the digits of integer bases. If $(d_k)_b = [b]$, then the value of d_k is $[b](b^k)$, which is more than b^{k+1} for all $b \notin \mathbb{Z}$. Therefore, some of this value could be expressed in the next largest place value with $d_{k+1} = 1$. This would decrease the value needed to be expressed in d_k from $[b](b^k)$ to

$$[b](b^k) - b^{k+1} = b^k([b] - b) \leq b^k.$$

So, the maximum digit we would ever need is $[b]$.

There are irrational number bases, which can more easily express certain irrational numbers than rational bases. One of the simplest examples of this is base- $\sqrt{2}$, which uses the digits 0 and

1 because $\lfloor \sqrt{2} \rfloor = 1$. Like negative and pure imaginary bases, the place values alternate what kind of number they are. Every even place value is rational and every odd one is irrational. Hence, we could express every integer by using just the even place values, which would essentially just be using binary. As an example, $3 + \frac{\sqrt{2}}{2} = 101.1_{\sqrt{2}}$.

Another irrational base system is that uses the golden ratio as its base. Also sometimes called the *tau system*, or base- $(\frac{1+\sqrt{5}}{2})$, or base- ϕ , or base- τ , or maybe even phinary, it has some unique properties. This system uses the digits 0 and 1, as $\lfloor \phi \rfloor = 1$. Perhaps surprisingly, the system $(\phi, \{0, 1\})$ is almost complete. Base- ϕ on its own has two unique representations for all integers. Since

$$\phi^n = \phi^{n-1} + \phi^{n-2},$$

the sequence 11_{ϕ} is equivalent to the sequence 100_{ϕ} , and so we will make the decision to always rewrite any sequence of digits 11 as a 100 for a number's standard representation.

The ability to rewrite two ones in a row as a single one in the next highest place value and vice versa is a very helpful tool when doing arithmetic in base- ϕ . For example, if we do $(101 + 11)_{\phi}$:

$$101_{\phi} + 11_{\phi} = 100_{\phi} + 1_{\phi} + 10_{\phi} + 1_{\phi} = 110_{\phi} + 1_{\phi} + 1_{\phi} = 1000_{\phi} + 1_{\phi} + 1_{\phi}$$

Now, we cannot add the two ones together to get a two or by carrying anything, so instead, we can rewrite one of them as 0.11_{ϕ} .

$$= 1000_{\phi} + 1_{\phi} + 0.11_{\phi} = 1001.11_{\phi} = 1010.01_{\phi}$$

So, $(101 + 11)_{\phi} = 1010.01_{\phi}$

If we list some of the powers of ϕ in terms of ϕ :

$$\phi^3 = 2\phi + 1$$

$$\phi^2 = \phi + 1$$

$$\phi^1 = \phi$$

$$\phi^0 = 1$$

$$\phi^{-1} = \phi - 1$$

$$\phi^{-2} = -\phi + 2$$

$$\phi^{-3} = 2\phi - 3$$

It is easier to see how we would form integers.

To form a 2, we would need the ϕ terms of its digits to sum to 0 and constant terms to sum to 2. Luckily, $\phi^1 + \phi^{-2}$ does just this. So, $2 = 10.01_{\phi}$.

Similarly, $3 = \phi^2 + \phi^{-2} = 100.01_{\phi}$, $4 = \phi^2 + \phi^0 + \phi^{-2} = 101.01_{\phi}$, etc.

We can also use other irrational numbers, like π and e , as bases, but these systems would not be as complete or interesting as base- ϕ .

7 Aspects of a Good Positional System

First, let us establish that, for everyday use, positive integer bases are the best, as any other type of base would be more convoluted and difficult to deal with.

While there is no universal rating system for bases, there are two major factors that determine the usefulness of a number base. One is how large it is: we want number bases to be relatively small so that we do not have to use and interpret a plethora of different digits but also not too small so that we do not need too many digits to express individual numbers. The second is how many factors the base has.

7.1 Divisibility in Different Bases

Divisibility is very important because if a base b is divisible by an integer n , then $\frac{1}{n}$ can be easily expressed in base- b . For example, $7 \mid 14$ and $\frac{1}{7} = 0.2_{14}$, just like how $\frac{1}{5} = 0.2_{10}$. Also, it is much easier to tell divisibility by n in base- b than in other bases not divisible by n . For example, in base-8, any integer with $d_0 \in \{0, 4\}$ is divisible by 4.

Therefore, in general, we want bases to have as many factors as possible given its relative size. It is important that these factors are small and/or prime, since smaller prime numbers are more common in the factorizations of numbers than larger ones, and smaller factors prevent the base from becoming too large. So, a very important aspect of a base is its prime factorization. How well fractions are expressed in a base (we want radix expressions that are short and that terminate) is also important.

In any base b , there are useful divisibility rules for both $b + 1$ and $b - 1$, like those of 9 and 11 in base-10.

An integer a in base- b can be expressed with digits d_k as $d_n \dots d_2 d_1 d_0$ where n is a nonnegative integer. So,

$$\begin{aligned} a &= d_n b^n + \dots + d_2 b^2 + d_1 b + d_0 \\ &= (d_n + d_n(b^n - 1)) + \dots + (d_2 + d_2(b^2 - 1)) + (d_1 + d_1(b - 1)) + d_0 \\ &= (d_n(b^n - 1) + \dots + d_2(b^2 - 1) + d_1(b - 1)) + (d_n \dots + d_2 + d_1 + d_0) \\ &= (d_n(b - 1)(b^{n-1} + b^{n-2} + \dots + b^2 + b + 1) + \dots + d_2(b - 1)(b + 1) + d_1(b - 1)) + (d_n \dots + d_2 + d_1 + d_0) \\ &= (b - 1)(d_n(b^{n-1} + b^{n-2} + \dots + b^2 + b + 1) + \dots + d_2(b + 1) + d_1) + (d_n \dots + d_2 + d_1 + d_0) \end{aligned}$$

Therefore, a is divisible by $b - 1$ if and only if $(d_n \dots + d_2 + d_1 + d_0)$, or the sum of the digits, is divisible by $b - 1$. This is the divisibility rule for 9 in base-10.

Now, for a divisibility rule for $b + 1$:
Let n be an odd nonnegative integer.

$$\begin{aligned} a &= d_n b^n + \dots + d_2 b^2 + d_1 b + d_0 \\ &= d_n(b^n + 1) - d_n + \dots + d_2(b^2 - 1) + d_2 + d_1(b + 1) - d_1 + d_0 \\ &= (d_n(b^n + 1) + \dots + d_2(b^2 - 1) + d_1(b + 1)) + (-d_n + \dots + d_2 - d_1 + d_0) \\ &= (d_n(b + 1)(b^{n-1} - b^{n-2} + \dots + b^2 - b + 1) + d_2(b + 1)(b - 1) + d_1(b + 1)) + (-d_n + \dots + d_2 - d_1 + d_0) \\ &= (b + 1)(d_n(b^{n-1} - b^{n-2} + \dots + b^2 - b + 1) + d_2(b - 1) + d_1) + (-d_n + \dots + d_2 - d_1 + d_0) \end{aligned}$$

Therefore, a is divisible by $b + 1$ if and only if $(-d_n + \cdots + d_2 - d_1 + d_0)$ is divisible by $b + 1$. This is the divisibility rule that we have for 11 in base-10.

As a side note, these rules are switched for negative bases, that is, a is divisible by $b - 1$ in base- $(-b)$ if $(-d_n + \cdots + d_2 - d_1 + d_0)$ is divisible by $b - 1$, and a is divisible by $b + 1$ in base- $(-b)$ if $(d_n + \cdots + d_2 + d_1 + d_0)$ is divisible by $b + 1$. For example, a number is divisible by 9 in negadecimal if the alternating sum of its digits is divisible by 9, and is divisible by 11 if the sum of its digits is divisible by 11. The reason for this is that every other place value in base- $(-b)$ is the additive inverse of the corresponding place value in base- b , so the alternating sum becomes the sum and the sum becomes the alternating sum.

A more general rule: if c is a factor of b , a number is divisible by c in base- b if d_0 is a multiple of c .

Like for 4 and 8 in base-10, there are useful divisibility rules in any base for all of the multiples of the base's factors.

Also, in any base, we can generate divisibility rules for primes that do not have a particularly nice relationship with the base, like that of 7 in base-10. We can also use the divisibility rules of two coprimes in combination for a way to tell divisibility by their product, like 6's rule in base-10 that uses divisibility by 2 and 3.

7.2 Advantages of Dozenal and Heximal

Although a societal shift to another number system would be very challenging and costly and not worth it, there are number bases, like base-12, also known as dozenal or duodecimal, and base-6, also known as heximal, seximal, or senary, that have advantages over our standardized base-10.

Although dozenal requires slightly more numerals than decimal (let χ be the digit for 10 and ϵ be the digit for 11 in dozenal), dozenal is considered the better of the two. This is thanks to the two-factor advantage of 12 over 10. Not counting themselves and 1, 10 has two factors and 12 has four.

However, heximal is also considered to be better than base-10 because it requires four less numerals than base-10 but has the same number of factors as 10.

To compare these three bases, we will compare the representations of fractions in each of them:

Of all of them, heximal appears to have the simplest representations of these eight fractions. However, whether heximal or duodecimal is considered better depends on what aspects of a base we are prioritizing.

8 The Uses of Some Bases

8.1 Binary

Binary is used to store and transmit information in digital circuits and computers because of its bipolar nature. It only uses two digits, and each place value can represent an on-and-off switch,

	Decimal	Duodecimal	Heximal
1/2	0.5	0.6	0.3
1/3	$0.\bar{3}$	0.4	0.2
1/4	0.25	0.3	0.13
1/5	0.2	$0.\overline{2497}$	$0.\bar{1}$
1/6	$0.1\bar{6}$	0.2	0.1
1/7	$0.14285\bar{7}$	$0.186\bar{\chi}$	$0.\overline{05}$
1/8	0.125	0.16	0.043
1/9	$0.\bar{1}$	0.14	0.04

Figure 4: Comparing fractions in different bases

with 0 typically representing off and 1 representing on. 0 results in a low, or false, voltage, while 1 results in a high, or true, voltage. Arithmetic in base-2 is also very simple. When adding in the same place value, a $0 + 0$ results in a 0, a $0 + 1$ results in a 1, and a $1 + 1$ results in a 1 in the place to the left and a 0 in the current place.

8.2 Hexadecimal

We also use hexadecimal (base-16) in computer science because it is a good middle ground between binary and decimal. Since 16 is a power of 2, computers can easily convert binary numbers into hexadecimal, and hexadecimal is easier for humans to read and interpret compared to binary because it more closely resembles decimal. Number representations in hexadecimal are much shorter than those in binary (4 times shorter, since $16 = 2^4$), and, to people, binary can appear like a meaningless string of 1s and 0s.

8.3 Quater-Imaginary

Computers use complex numbers in many different ways, and the arithmetic of complex numbers is important to several of the functions of a computer. Because of this, quater-imaginary base (base- $2i$) has the potential to be a more efficient number system than binary for computers. Quater-imaginary can represent negative numbers without a sign. Furthermore, it can represent all complex numbers, not just real and pure imaginary numbers, as just one part, whereas real bases need both a real and imaginary part. So, arithmetic with complex numbers can be done more efficiently in quater-imaginary than in binary, especially in regards to multiplication. In real bases, multiplying two complex numbers $a + bi$ and $c + di$ where $a, b, c, d \neq 0$ requires four multiplications and three additions:

$$(a + bi)(c + di) = ac + a(di) + c(bi) + (bi)(di)$$

In base- $2i$, this process only requires one multiplication.

9 References

“A Brief History of Numbers: How 0-9 Were Invented.” Casio Education, Aug 2018.
<https://education.casio.co.uk/blog/a-brief-history-of-numbers>.

“Additive, Multiplicative, and Ciphred Systems of Numeration.”

<https://www.hcusd2.org/vimages/shared/vnews/stories/53bd4762d3b99/4.1%20%20Additive,%20Multiplicative,%20and%20Ciphred%20Systems%20of%20Numeration%20-%20filled%20in.pdf>

Bergman, George. "A Number System with an Irrational Base." *Mathematics Magazine*, Vol. 31, No. 2. December, 1957.

https://math.berkeley.edu/~gbergman/papers/base_tau.pdf

“Ancient Chinese Mathematics.” <https://www.math.uci.edu/~ndonalds/math184/china.pdf>.

“Fractional and Non-Integer Number Bases.” Brilliant, retrieved June 14, 2023.

<https://brilliant.org/wiki/fractional-number-base/>.

Gilbert, William J. “Arithmetic in Complex Bases.” University of Waterloo, March 1984.

<https://www.math.uwaterloo.ca/~wgilbert/Research/ArithCxBases.pdf>.

Gomony, Manil Dev, and Souradip Sarkar. “Quater-Imaginary Base for Complex Number Arithmetic Circuits.” *Hardware Platforms Research: Nokia Bell Labs*, 2018.

<https://past.date-conference.com/proceedings-archive/2018/pdf/0029.pdf>.

Grussenmeyer, Kristina., & Arikan, Beytullah. "History of Zero."

<http://webspace.ship.edu/msrenault/400%20Presentations/Presentation%203%20Zero.pdf>

Hanson, Marilee. “Roman Numerals” *English History*, May 10, 2022.

<https://englishhistory.net/romans/roman-numerals/>.

Jacobson, Roni. “60: Behind Every Second, Millenniums of History.”

The New York Times, July 8, 2013.

<https://www.nytimes.com/2013/07/09/science/60-behind-every-second-millenniums-of-history.html#:~:text=In%20Babylonian%20astronomy%2C%20a%20year,of%2060%20from%20the%20hour.>

Katai, I., & Szabo, J. “Canonical number systems for complex integers.” December 28, 1974.

<https://web.northeastern.edu/seigen/11Magic/Binary/Complex/Canonical%20Number%20Systems%20for%20Complex%20Integers%20%20Katai%20%20Szabo.pdf>

MacTutor, November 2000. <https://mathshistory.st-andrews.ac.uk/HistTopics/Zero/>.

O’Connor, J. J., & Robertson, E. F. “Babylonian Numerals.” *Maths History*, December 2000.

https://mathshistory.st-andrews.ac.uk/HistTopics/Babylonian_numerals/.

O’Connor, J. J., & Robertson, E. F. "The Arabic numeral system." January, 2001.

https://mathshistory.st-andrews.ac.uk/HistTopics/Arabic_numerals/

“Phi Numeral System.” *Phi numeral system - OeisWiki*, May 9, 2021.

https://oeis.org/wiki/Phi_numeral_system.

Piché, Daniel G. “Complex Bases, Number Systems and Their Application to Fractal Wavelet Image Coding.” University of Waterloo, 2002.
<https://www.collectionscanada.gc.ca/obj/s4/f2/dsk3/OWTU/TC-OWTU-243.pdf>

“Radix notation.” nLab. <https://ncatlab.org/nlab/show/radix+notation>.

Savas, Niko. “Why do we use hexadecimal?” Medium, Sep 23, 2016.
<https://medium.com/@savas/why-do-we-use-hexadecimal-d6d80b56f026>.

Storr, Wayne. “Binary Numbers and the Binary Number System.” Basic Electronics Tutorials, August 8, 2022.
https://www.electronics-tutorials.ws/binary/bin_1.html.